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30 JANUARY 1963
SUPERSEDING
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19 SEPTEMBER 1958

MILITARY SPECIFICATION

POWERED MAGNETIC MATERIAL, MOLDED, STABILITY OF (FOR CORES USED IN ELECTRICAL AND ELECTRONIC PARTS)

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the evaluation of the stability, under environmental conditions, of molded, powdered magnetic material for use as cores in such electrical and electronic parts as coils, transformers, etc. (see 6.1). For purposes of this specification, the term "magnetic material" is used herein when referring to the molded, powdered magnetic material.

1.2 Classification. Magnetic material shall be of the following classes and grades, as specified (see 6.2):

Class CL	—Maximum operating temperature of 150° C.
Class CC	—Maximum operating temperature of 200° C.
Class CH(. . .)[1]	—Maximum operating temperature of 225° C. or above, as rated by the manufacturer.
Grade M	—Moisture resistant.
Grade U	—For use in sealed assemblies.

[1] This class designation is completed by inserting in the parentheses the temperature rating by the manufacturer. The temperature rating shall be assigned in steps of 25° C. above 200° C.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Federal Test Method Std. No. 406—Methods of Testing

MILITARY

MIL–STD–202 —Test Method for Electronic and Electrical Component Parts

(Copies of standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

Metal Powder Industries Federation

MPI33–62—Recommended Method for Determination of Toroidal Core Inductance and Permeability

FSC 5950

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(Applications for copies should be addressed to the Metal Powder Industries Federation, 60 East 42d, St., New York 17, N. Y.)

3. REQUIREMENTS

3.1 Qualification. Magnetic material furnished under this specification shall be a product which has been tested, and has passed the qualification inspection specified in 4.4, and has been listed on or approved for listing on the applicable Qualified Products List (see 6.3).

3.2 Material. The material shall be a pow-

TABLE I. *Property values for molded magnetic material*

Property to be tested	Method paragraph	Conditioning procedure (See 4.6.1.3)	Value required					
			Class CL		Class CC		Class CH (____)	
			Grade M	Grade U	Grade M	Grade U	Grade M	Grade U
Density	4.6.3.1	None	Value shall be noted and recorded					
Diameter	4.6.3.2	None	Gage shall be noted and recorded					
		M	(1)	.....	(1)	.....	(1)	.....
		U	.....	(1)	.....	(1)	.....	(1)
		CL	(1)	(1)	.....	.....	.....	.....
		CC	.....	.....	(1)	(1)	.....	.....
		CH (....)	.....	.....	.....	.....	(1)	(1)
Quality factor (Q)	4.6.3.3	None	Value shall be noted and recorded					
		M	(2)	.....	(2)	.....	(2)	.....
		U	.....	(2)	.....	(2)	.....	(2)
		CL	(2)	(2)	.....	.....	.....	.....
		CC	.....	.....	(2)	(2)	.....	.....
		CH (....)	.....	.....	.....	.....	(2)	(2)
Resonant capacity (C)	4.6.3.4	None	Value shall be noted and recorded (3)					
		M	(2, 3)	.....	(2, 3)	.....	(2, 3)	.....
		U	.....	(2, 3)	.....	(2, 3)	.....	(2, 3)
		CL	(2, 3)	(2, 3)	.....	.....	.....	.....
		CC	.....	.....	(2, 3)	(2, 3)	.....	.....
		CH (....)	.....	.....	.....	.....	(2, 3)	(2, 3)
Dielectric withstanding voltage, volts, minimum.	4.6.3.5	M	50	.....	50	.....	50	.....
		U	.....	50	.....	50	.....	50
		CL	50	50	.....	.....	.....	.....
		CC	.....	.....	50	50	.....	.....
		CH (....)	.....	.....	.....	.....	50	50
Mechanical strength: For thermosetting binders, pounds, minimum.	4.6.3.6.1	M	1,000	.....	1,000	.....	1,000	.....
		U	.....	1,000	.....	1,000	.....	1,000
		CL	1,000	1,000	.....	.....	.....	.....
		CC	.....	.....	1,000	1,000	.....	.....
		CH (....)	.....	.....	.....	.....	1,000	1,000
For thermoplastic binders.	4.6.3.6.2	M	(4)	.....	(4)	.....	(4)	.....
		U	.....	(4)	.....	(4)	.....	(4)
		CL	(4)	(4)	.....	.....	.....	.....
		CC	.....	.....	(4)	(4)	.....	.....
		CH (....)	.....	.....	.....	.....	(4)	(4)

[1] Each conditioned specimen shall pass freely through the same or the next larger gage, as noted before conditioning.

[2] The value for the test coil (see appendix) with the conditioned specimen inserted shall not deviate by more than 10 percent from the value obtained for the same test coil with the same specimen inserted before conditioning.

[3] No single value for the test coil (see appendix) with a specimen inserted shall differ by more than 5 percent from the average of all the values for all the specimens of a set.

[4] Specimen shall not crack or fracture.

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dered magnetic molding material containing a binder and such other ingredients as are necessary, and shall be so formulated that when molded into specimen cores, such cores shall meet the requirements of this specification.

3.3 **Stability.** Each specimen tested shall meet the requirements of table I.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 *Supplier.* The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1.1 *Test equipment and inspection facilities.* Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 Classification of inspection. The examination and testing of magnetic material shall be classified as follows:

(a) Qualification inspection (see 4.4).
(b) Acceptance inspection (see 4.5).
1. Inspection of product for delivery (see 4.5.1).

4.3 Inspection conditions. Unless otherwise specified herein, all inspection shall be made at room ambient temperature, relative humidity, and pressure.

4.4 Qualification inspection. Qualification inspection will be performed at a laboratory designated by the Government (see 6.3).

4.4.1 *Sample.* Nine specimens (see 4.6.1), representative of each of the material blends and processes used in production, shall be submitted for qualification inspection (see 4.4.3).

4.4.2 *Test routine.* Specimens will be subjected to the qualification inspection specified in table II, in the order shown. All specimens will be subjected to the inspection of group I. Six of these specimens will then be divided equally into two groups of three specimens each. The six specimens will then be conditioned as specified in table II and subjected to the inspection for their particular group. The remaining three specimens will be set aside and used as references.

4.4.3 *Test data.* The supplier shall furnish, with the specimens, test data (determined from at least three specimens) indicating that the magnetic material meets the requirements of this specification. In addition, he shall submit information on initial permeability, measured in accordance with MPIF Standard 33–62, and temperature coefficient as determined on a toroid, with no conditioning, having an outside diameter of 0.932 to 0.942 inch, inclusive, an inside diameter of 0.620 to 0.632 inch, inclusive, and a height (length) of 0.156 inch ±20 percent.

TABLE II. *Qualification inspection*

Examination or test	Requirement table	Method paragraph
Group I (all specimens) (With no conditioning)		
Density		4.6.3.1
Diameter		4.6.3.2
Quality factor (Q)		4.6.3.3
Resonant capacity (C)		4.6.3.4
Group II (3 specimens) (After condition M or U, as applicable)		
Diameter	Table I	4.6.3.2
Quality factor (Q)		4.6.3.3
Resonant capacity (C)		4.6.3.4
Dielectric withstanding voltage		4.6.3.5
Mechanical strength		4.6.3.6
Group III (3 specimens) (After condition CL, CC, or CH (....), as applicable).		
Diameter		4.6.3.2
Quality factor (Q)		4.6.3.3
Resonant capacity (C)		4.6.3.4
Dielectric withstanding voltage		4.6.3.5
Mechanical strength		4.6.3.6

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4.4.4 *Failure.* Failure in any of the examinations or tests will be cause for refusal to grant qualification.

4.4.5 *Extent of qualification.* Qualification of the specimens submitted will be the basis for qualification of all magnetic material with a density up to and including the highest density of the specimens submitted.

4.5 **Acceptance inspection.** Acceptance inspection shall be performed upon powdered magnetic material for cores in the form of finished cores in the component or part whose specification or drawing references this specification. The sampling and inspection procedures shall be as specified therein. When not specified, sampling and inspection procedures shall be in accordance with 4.5.1 to 4.5.1.4, inclusive.

4.5.1 *Inspection of product for delivery.* Insection of product for delivery shall consist of batch inspection (see 4.5.1.2) and periodic batch inspection (see 4.5.1.3).

4.5.1.1 *Batch.* A batch, as far as practicable, shall consist of all the magnetic material of the same class and grade, produced under essentially the same conditions, and offered for inspection at one time.

4.5.1.2 *Batch inspection.* Batch inspection shall consist of the examinations and tests specified in table III, in the order shown for each group.

TABLE III. *Batch inspection*

Examination or test	Requirement table	Method paragraph
Group 1 (all specimens) (With no conditioning)	Table I	
Density		4.6.3.1
Diameter		4.6.3.2
Quality factor (Q)		4.6.3.3
Resonant capacity (C)		4.6.3.4
Group 2 (all specimens) (After condition CL, CC, or CH (....), as applicable).		
Diameter		4.6.3.2
Quality factor (Q)		4.6.3.3
Resonant capacity (C)		4.6.3.4
Dielectric withstanding voltage		4.6.3.5

4.5.1.2.1 *Sampling plan.* Batch inspection shall be made on each batch and shall be the basis for acceptance or rejection of the batch. Three specimens (see 4.6.1) shall be prepared. If any specimen fails to pass any of the examinations or tests in table III, the batch shall be considered to have failed.

4.5.1.3 *Periodic batch inspection.* Periodic batch inspection shall consist of the examinations and tests specified in table IV. A different set of specimens shall be selected for each group and shall be tested in the order shown. Shipment shall not be held up pending results of the inspection.

TABLE IV. *Periodic batch inspection*

Examination or test	Requirement table	Method paragraph
Group 1 (3 specimens) (After condition M or U, as applicable).	Table I	
Diameter[1]		4.6.3.2
Quality factor (Q)[1]		4.6.3.3
Resonant capacity (C)[1]		4.6.3.4
Dielectric withstanding voltage		4.6.3.5
Mechanical strength		4.6.3.6
Group 2 (3 specimens) (After condition CL, CC, or CH (....), as applicable.		
Mechanical strength		4.6.3.6

[1] The initial (unconditioned) value for these examinations and tests shall be that determined on the same specimen during batch inspection (see 4.5.1.2) of the batch on which periodic batch inspection is made.

4.5.1.3.1 *Sampling plan.* Periodic batch inspection shall be made on the first batch and on every twentieth batch thereafter, or once every 2 years, whichever is more frequent. Six specimens (see 4.6.1) shall be prepared. If any specimen fails to pass any of the examinations or test in table IV, the batch shall be considered to have failed.

4.5.1.4 *Noncompliance.* If a batch fails to pass periodic batch inspection, the supplier shall take corrective action on the materials or process, or both, as warranted, and on all batches which can be corrected and which were manufactured under essentially the same conditions, with essentially the same material, processes, etc, and which are considered subject to the same failure. Accept-

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ance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, periodic batch inspection shall be repeated on additional specimens (all inspection, or the inspection which the original specimens failed, at the option of the Government). Batch inspection may be reinstituted; however, final acceptance shall be withheld until the periodic batch reinspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and the corrective action taken shall be furnished to the contracting officer.

4.6 Methods of examination and test.

4.6.1 *Specimens.*

4.6.1.1 *Sampling.* The sample of magnetic material from which specimens are to be molded shall be so selected as to be representative of the batch being inspected by taking samples from the top, middle, and bottom of the batch, thoroughly mixing and consolidating the samples, and reducing the consolidated sample by quartering.

4.6.1.2 *Form and preparation.* Specimens shall be cylinders 0.3674±0.0025 inch in diameter by 0.750±0.015 inch long, molded into shape by methods which are representative of production.

4.6.1.3 *Conditioning.*

4.6.1.3.1 *Condition M.* Condition M shall consist of the following procedure:

(a) Temperature cycling in accordance with method 102, test condition D, of MIL–STD–202. Measurements of Quality Factor (Q) and Resonant Capacity (C) shall be made both before and after temperature cycling.

(b) Moisture resistance in accordance with method 106 of MIL–STD–202. The following exceptions shall apply:

1. Mounting of specimens—Not applicable.
2. Subcycle—During step 7b, the vibration shall be omitted.
3. Loading voltage—Not applicable.

(c) Cold storage at −65°±2° C for 72 +1 hours.

(d) Stabilization at room ambient temperature before test.

4.6.1.3.2 *Condition U.* Condition U shall consist of the same procedure as condition M, except that item (b), moisture resistance, shall be omitted.

4.6.1.3.3 *Conditions CL, CC, and CH (...).* Conditions CL, CC, and CH(...) shall consist of storage of the specimens in a circulating-air oven for 96+1 hours at temperatures, respectively, of 150°±2° C, 200°±2° C, and 225°±2° C or above (as specified by the manufacturer). The specimens shall then be stabilized at room ambient temperature before test.

4.6.2 *Determination of test frequency.* Three unconditioned specimens shall be subjected to the quality-factor (Q) test (see 4.6.3.3) at frequencies of 150 kilocycles per second (kc/sec), 800 kc/sec, 1.50 megacycles per second (mcs), 15 mcs, and 60 mcs. The frequency at which the maximum quality factor (Q) occurs shall be used as the test frequency.

4.6.3 *Apparatus and procedure.*

4.6.3.1 *Density.* Density shall be determined in accordance with method 5012 of Fed. Test Method Standard No. 406, except that the weight shall be divided by the volume.

4.6.3.2 *Diameter.*

4.6.3.2.1 *Gages.* Seven diameter gages having an internal diameter, $^{+0.0002}_{-0.0000}$ inch, of 0.3650, 0.3660, 0.3670, 0.3680, 0.3690, 0.3700, and 0.3710 inch, respectively, shall be used.

4.6.3.2.2 *Procedure.* The specimen shall be passed through the gages, and the smallest gage through which it passes freely shall be determined.

4.6.3.3 *Quality factor (Q).* A specimen, dimensionally acceptable shall be inserted fully into the test-coil assembly (see appendix) and the quality factor (Q) shall then be determined in accordance with method 306 of MIL–STD–202, at the test frequency (see 4.6.2).

4.6.3.4 *Resonant capacity (C).* A specimen,

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dimensionally acceptable, shall be inserted fully into the test-coil assembly (see appendix), and the capacity necessary to resonate the coil at the test frequency (see 4.6.2) shall then be determined.

4.6.3.5 *Dielectric withstanding voltage.* Specimens shall be tested in accordance with method 301 of MIL-STD-202. The following details and exception shall apply:

(a) Magnitude and nature of test voltage—Test voltage shall be raised from 0 to 50 volts direct current at a rate of approximately 2½ volts per second.

(b) Duration of application of test voltage—Held at 50 volts for 60 seconds, during which time the current shall not exceed 50 milliamperes.

(c) Point of application of test voltage—The electrodes shall be in intimate contact with the end planes of the specimen. The end planes shall be flat, parallel, and free of coating, surface insulating material, or any other foreign material.

(a) Voltage at which failure occurs—If failure occurs before 50 volts is reached, the voltage at which failure occurs shall be noted and recorded.

4.6.3.6 *Mechanical strength.*

4.6.3.6.1 *For thermosetting binders.* Mechanical strength shall be determined in accordance with method 1022 of Fed. Test Method Standard No. 406. The load shall be applied axially.

4.6.3.6.2 *For thermoplastic binders.* Mechanical strength shall be determined in accordance with 4.6.3.6.2.1 and 4.6.3.6.2.2.

4.6.3.6.2.1 *Apparatus.* A steel ball weighing ½ pound shall be used. The steel ball shall be held and released by a rigidly supported magnetic device designed to drop the ball on the exact center of the specimen, located on a steelplate 15 inches by 24 inches by 1 inch.

4.6.3.6.2.2 *Procedure.* The specimen shall be laced with its axis perpendicular to the steelplate and located so that when the ball is dropped a vertical distance of 1 foot, it will strike the center of the specimen.

5. PREPARATION FOR DELIVERY

5.1 Not applicable to this specification.

6. NOTES

6.1 Intended use. The magnetic material covered by this specification is intended for use in applications where stability of electrical and magnetic characteristics under adverse environmental conditions is a prime requisite. Such stability is evaluated on standard cylindrical specimen cores; in addition, minimum requirements for mechanical and electrical strength are established. Specific performance of cores, however, is not covered herein; such requirements are incorporated in the performance specification for the coil, transformer, etc.

6.1.1 *Classes CL, CC, and CH(. . .).* These classes are intended for use at elevated temperatures as indicated by the class, either in sealed or unsealed assemblies as indicated by the grade.

6.1.2 *Grade M.* Grade M is intended for use in unsealed assemblies which are subjected to adverse humidity conditions.

6.1.3 *Grade U.* Grade U is intended for use in sealed assemblies, or in applications under normal atmospheric conditions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Class and grade desired (see 1.2).

6.3 Qualification. With respect to products qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government,

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tested for qualification, in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is Signal Corps; however, information pertaining to qualification of products may be obtained from the Defense Electronics Supply Center, 1507 Wilmington pike, Dayton 20, Ohio.

Custodians:
Army—Electronic Command
Navy—Ships
AF—AFSC

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APPENDIX

DESIGN AND CONSTRUCTION OF TEST-COIL ASSEMBLY

10. SCOPE

10.1 This appendix describes the design and construction of the test-coil assembly used for the determination of quality factor (Q) and resonant capacity (C).

20. APPLICABLE DOCUMENTS

20.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this appendix to the extent specified herein.

SPECIFICATIONS

FEDERAL

L–M–500 —Molding Plastic, Acrylic
QQ–B–613 —Brass, Leaded and Non-Leaded; Plate, Rolled Bar, Sheet and Strip

MILITARY

MIL–P–79 —Plastic Rods and Tubes, Thermosetting, Laminated
MIL–W–583 —Wire, Magnet, Electrical
MIL–A–8576—Adhesive, Acrylic Monomer Base, for Acrylic Plastic

(Copies of specifications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

30. REQUIREMENTS

30.1 Material.

30.1.1 *Plastic.* Plastic material, except for the stop, shall conform to Specification L–M–500. Plastic material for the stop shall conform to type PBE of MIL–P–79.

30.1.2 *Metal.* Metal parts shall be brass conforming to Specification QQ–B–613.

30.1.3 *Wire.* Magnet wire shall conform to MIL–W–583 (see table V).

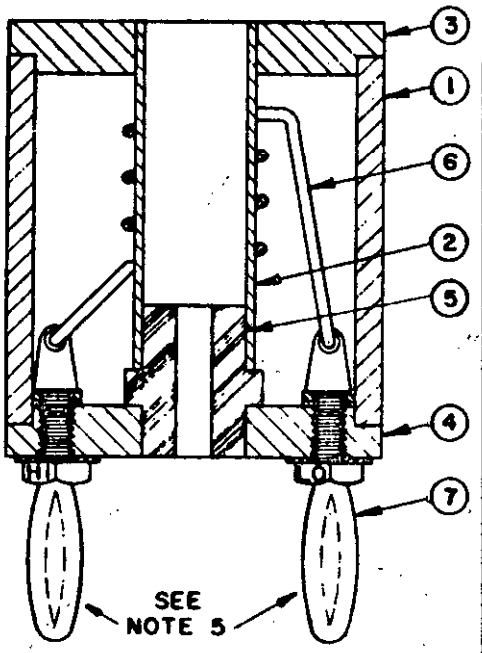
30.2 Design and construction. The test-coil assembly shall be of the design, construction, and physical dimensions specified on figure 1.

30.2.1 *Test-coil winding.* Test coils shall be wound as specified in table V. Tolerance on dimensions shall be $\pm 1/64$ inch.

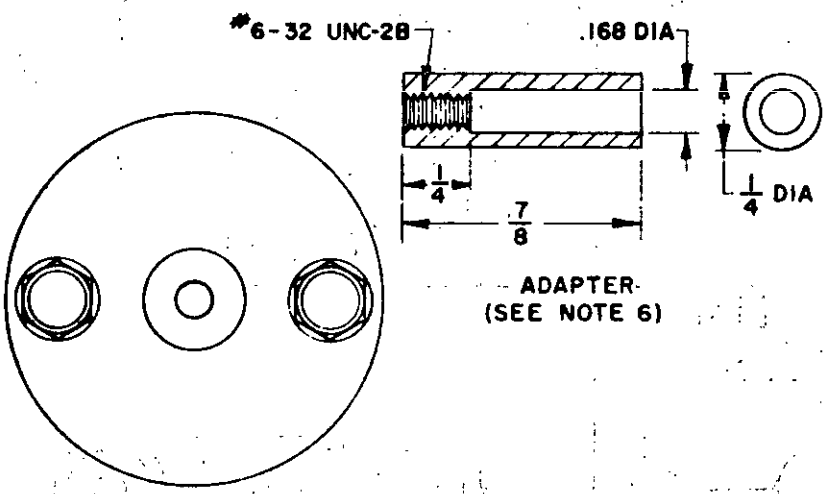
TABLE V. *Test coil*

Test frequency	Test-coil values		Winding information
	Capacity (uuf)	Quality factor (Q) (See 4.6.3.8)	
150 kc/sec.....	435	90, ±10 percent	Three universal-wound pies, each 11/64 inch ling, consisting of 180 to 200 turns of 6 strands of type TS magnet wire, in AWG size 42. Press pies close together to total length of 17/32 inch.
800 kc/sec.....	578	114, ±10 percent	Progressively wound solenoid 5/8 inch long, consisting of 104 to 120 turns of 15 strands of type TS magnet wire, in AWG size 44.
1.50 mcs.....	697	107, ±10 percent	Solenoid winding 5/8 inch long, consisting of 52½ to 56½ turns of 15 strands of type TS magnet wire, in AWG size 44.
15 mcs.....	330	149, ±10 percent	Space-wound solenoid 5/8 inch long, consisting of 6½ turns of type T magnet wire, in AWG size 18.
60 mcs.....	47	285, + infinity, —10 percent	Space-wound solenoid 5.8 inch long, consisting of 3½ turns of type T magnet wire, in AWG size 14.

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Item	Description	Material	Qty
1	Side housing	Acrylic, Type II, Class 2, per spec. L-M-500	1
2	Test coil form	Acrylic, Type II, Class 2, per Spec. L-M-500 or Phenolic-resin, paper-base type PBE, per Spec. MIL-P-79	1 1
3	Top housing	Acrylic, Type II, Class 2 per Spec. L-M-500	1
4	Base housing		
5	Stop	Phenolic-resin, paper-base type PBE, per Spec. MIL-P-79	1
6	Test-coil winding (see table V)	Magnet wire per Spec. MIL-W-583	1
7	Terminal	General Radio type 274-P, or equal, banana plug with #6-32 nut and solder clip. Nut to be 3/16 inch OD for Spanner wrench.	2
---	Adapter	Brass per Spec QQ-B-613	2 (if reqd)



1. All dimensions in inches.
2. Tolerances are ±1/64 on fractions.
3. Joints shall be securely bonded, where applicable, using adhesive conforming to Specification MIL-A-8576.
4. Terminals shall be marked "H1" and "LO".
5. For test frequencies of 150 kc/sec to 50 mcs, the spacing between terminals, item 7, and between meter terminals shall be 1.000 inch; for 60 mcs, the spacing shall be 0.750 inch.
6. Inside diameter of adapter shall be such as to fit terminals, item 7.

FIGURE 1. *Test-coil assembly.*

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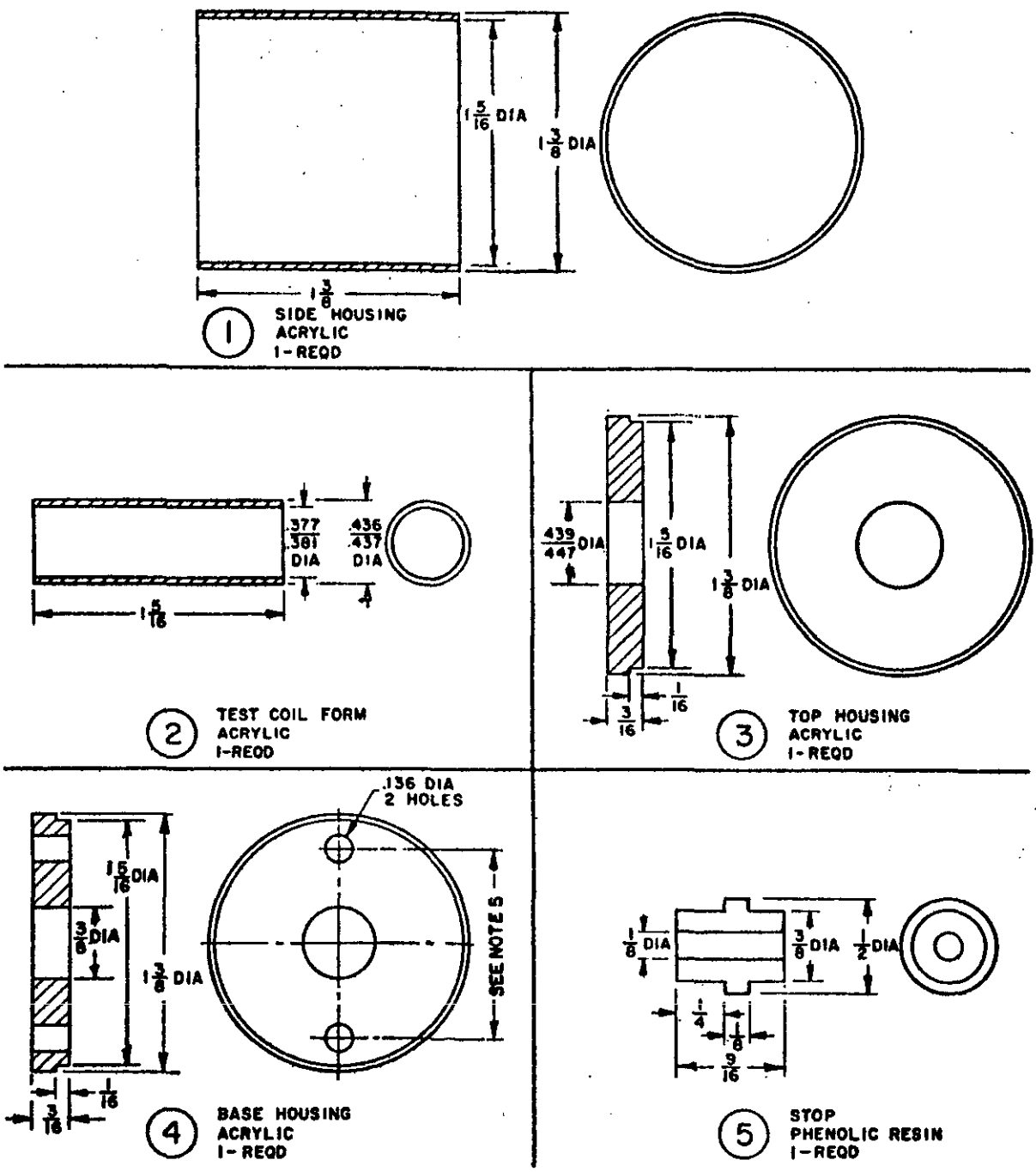


FIGURE 1.—Continued

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